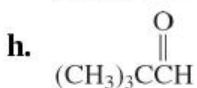
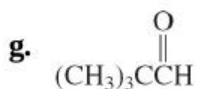
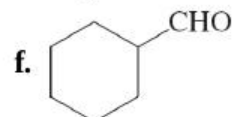
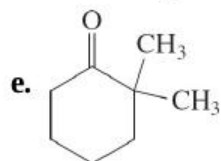
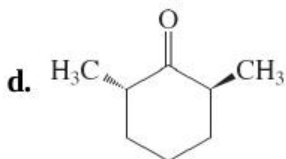
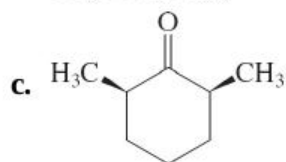


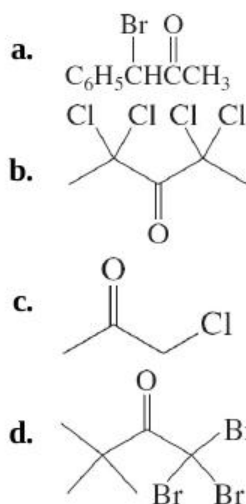
Problems

32. Underline the α -carbons and circle the α -hydrogens in each of the following structures.

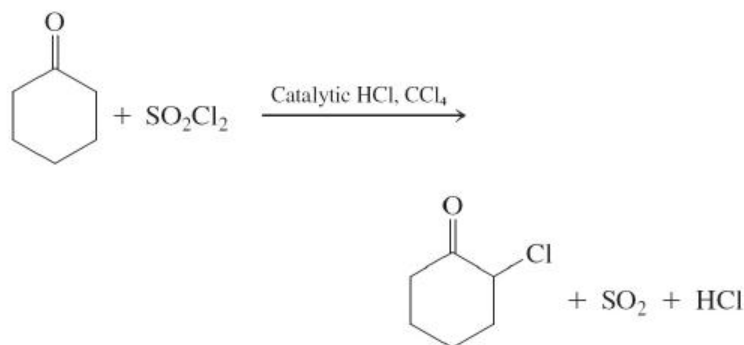


33. Write the structures of every enol and enolate ion that can arise from each of the carbonyl derivatives illustrated in [Problem 32](#).
34. What product(s) would form if each carbonyl compound in [Problem 32](#) were treated with (a) alkaline D_2O ; D_2O ; (b) 1 equivalent of Br_2 in acetic acid; (c) excess Cl_2 in aqueous base?
35. Describe the experimental conditions that would be best suited for the efficient synthesis of each of the following molecules from the

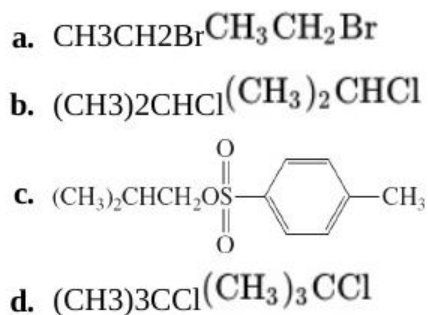
corresponding nonhalogenated ketone.



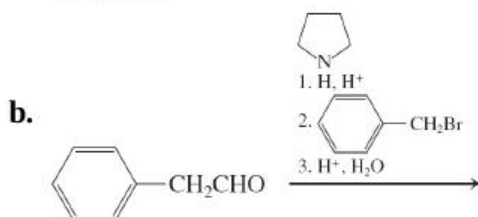
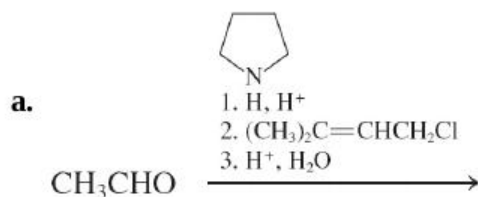
36. Propose a mechanism for the following reaction. (**Hint:** Take note of all of the products that are formed and base your answer on the mechanism for acid-catalyzed bromination of acetone shown in [Section 18-3](#).)



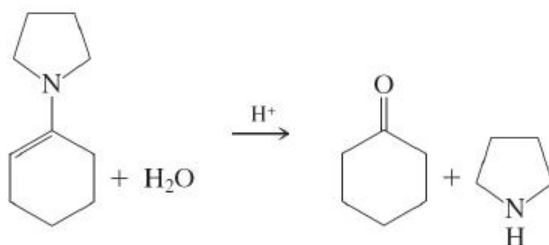
37. Give the product(s) that would be expected on reaction of 3-pentanone with 1 equivalent of LDA, followed by addition of 1 equivalent of



38. Give the product(s) of the following reaction sequences.

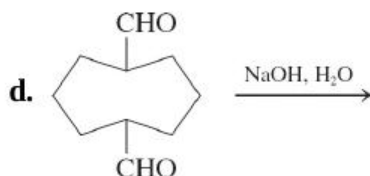
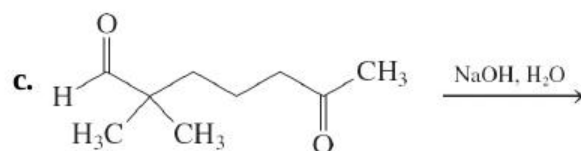
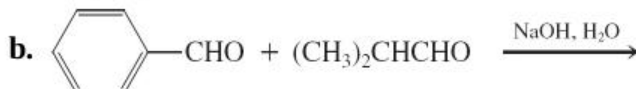
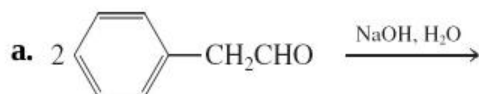


39. The problem of double compared with single alkylation of ketones by iodomethane and base is mentioned in [Section 18-4](#). Write a detailed mechanism showing how some double alkylation occurs even when only one equivalent each of the iodide and base is used. Suggest a reason why the enamine alkylation procedure solves this problem.
40. Would the use of an enamine instead of an enolate improve the likelihood of successful alkylation of a ketone by a secondary haloalkane?
41. Formulate a mechanism for the acid-catalyzed hydrolysis of the pyrrolidine enamine of cyclohexanone (shown below).

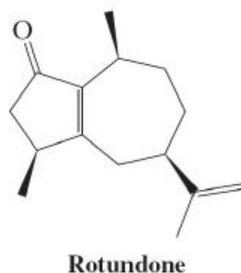


42. Write the structures of the aldol condensation products of (a) pentanal; (b) 3-methylbutanal; (c) cyclopentanone.
43. Write the structures of the expected major products of crossed aldol condensation at elevated temperature between excess benzaldehyde and (a) 1-phenylethanone (acetophenone—see [Section 17-1](#) for structure); (b) acetone; (c) 2,2-dimethylcyclopentanone.

44. Formulate a detailed mechanism for the reaction that you wrote in [Problem 43](#), part (c).
45. Give the likely products for each of the following aldol addition reactions.



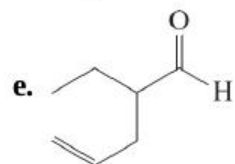
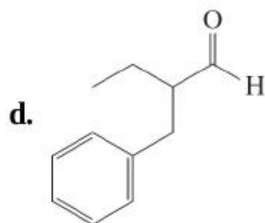
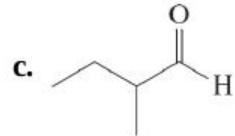
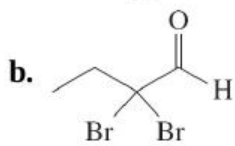
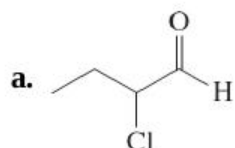
46. Rotundone (top of next column) is the natural product responsible for the peppery aroma in peppers, many herbs, and red wines ([p. 885](#)). What cyclic diketone will give rotundone upon intramolecular aldol condensation?

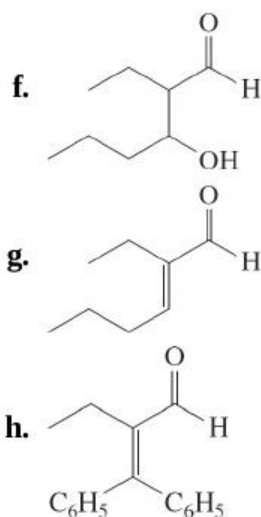


47. Write *all* possible products of the base-catalyzed crossed aldol reactions between each pair of reaction partners given below. (**Hint:** Multiple products are possible in every case; be sure to include thermodynamically unfavorable as well as favorable ones.)
- a. Butanal and acetaldehyde

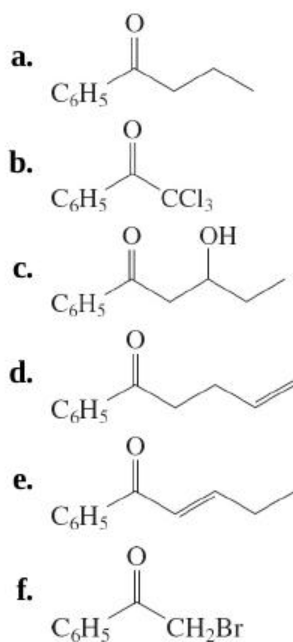
- b. 2,2-Dimethylpropanal and acetophenone
- c. Benzaldehyde and 2-butanone

48. For each of the three crossed aldol reactions described in [Problem 47](#), indicate which, if any, of the multiple possible products should predominate in the reaction mixture and explain why.
49. Aldol condensations may be catalyzed by acids. Suggest a mechanistic role for H^+ in the acid-catalyzed version. (**Hint:** Consider what kind of nucleophile might exist in acidic solution, where enolate ions are unlikely to be present.)
50. Reaction review. Without consulting the Reaction Road Maps on [pp. 852–853](#), suggest reagents to convert butanal into each of the following products.

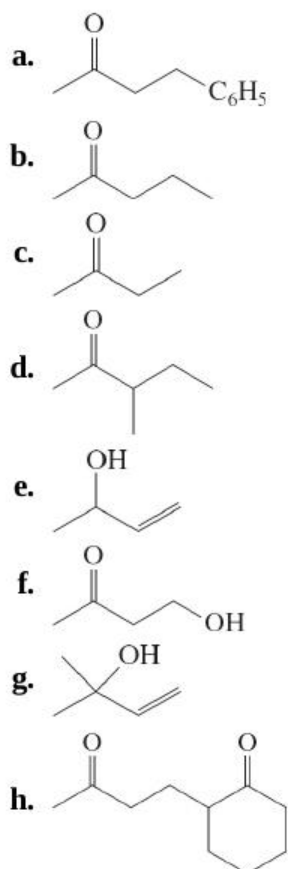




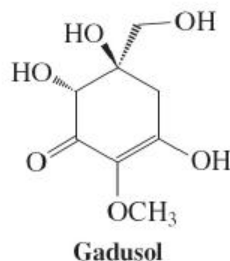
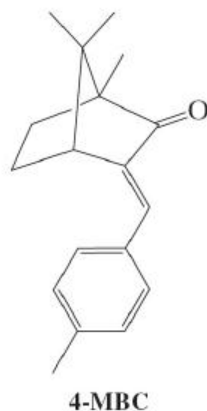
51. Reaction review II. Without consulting the Reaction Road Maps on [pp. 852–853](#), suggest reagents to convert acetophenone into each of the following substances.



52. Reaction review III. Without consulting the Reaction Road Maps on [pp. 852–853](#), suggest reagents to convert 3-buten-2-one into each of the following compounds.

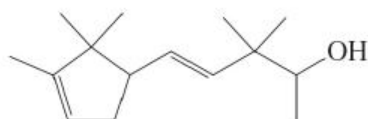


53. (a) A number of highly conjugated organic substances are ingredients of sunscreens. One of the more widely used is 4-methylbenzylidene camphor (4-MBC), whose structure is shown below. This compound is effective in absorbing so-called UV-B radiation (with λ_{max} values between 280 and 320 nm, responsible for most sunburns). Suggest a simple synthesis of this molecule using a crossed aldol condensation.



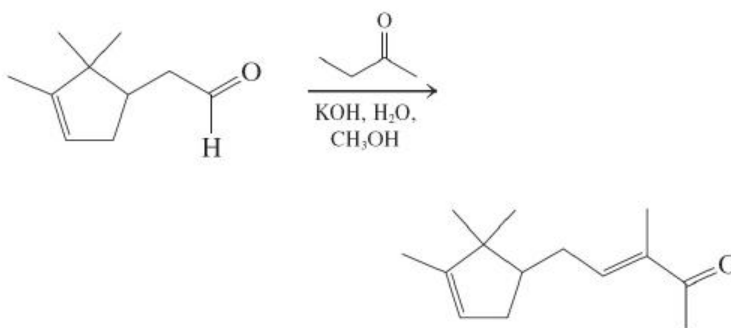
(b) The compound gadusol, above right (note the preference for the enol tautomer), was found in 2015 to be produced as a natural sun protectant by numerous organisms including fish and some vertebrates. The need for “green,” environmentally friendly sunscreens has generated interest in such biobased substances. What are the structural features of 4-MBC and gadusol that impart their ability to absorb ultraviolet light?

54. **CHALLENGE** The distillate from sandalwood is one of the oldest and most highly valued fragrances in perfumery. The natural oil is in short supply and, until recently, synthetic substitutes have been difficult to prepare. Polysantol (below) is the most successful of these substitutes.

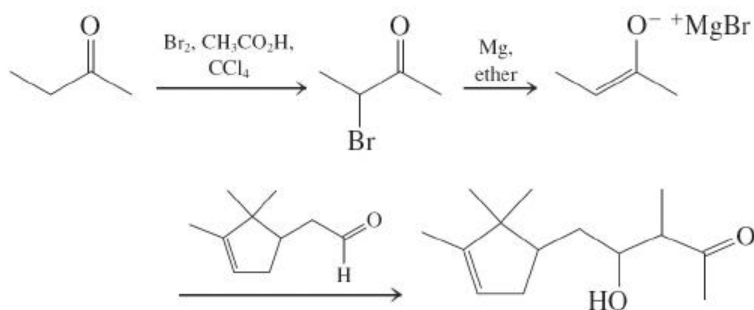


Polysantol

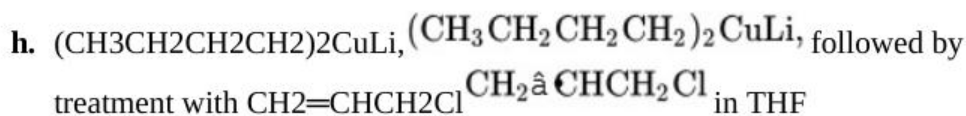
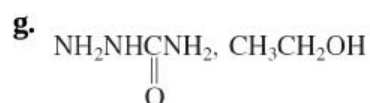
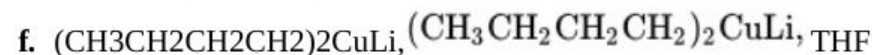
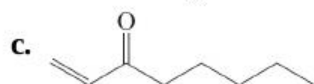
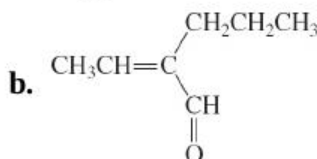
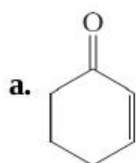
Its synthesis involves the aldol condensation shown below.



- This step, although usable, had a significant drawback that was responsible for its modest yield (60%). Identify this problem in detail.
- A procedure that avoids the conventional aldol condensation protocol was published more recently. A bromoketone is prepared and, upon reaction with Mg metal, gives a magnesium enolate. This enolate then reacts with the aldehyde selectively to give a hydroxyketone that may be dehydrated to the desired product. Discuss how this approach solves the problem outlined in part (a).

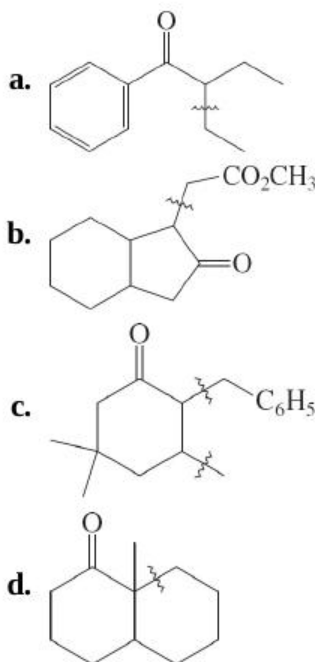


55. Write the expected major product of reaction of each of the carbonyl compounds (i)–(iii) with each of the reagents (a)–(h).

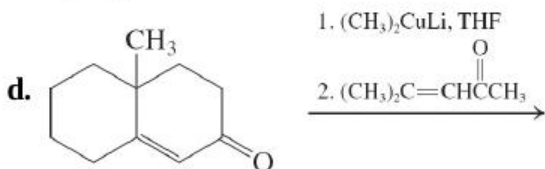
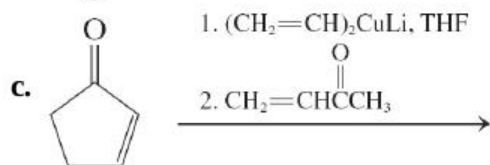
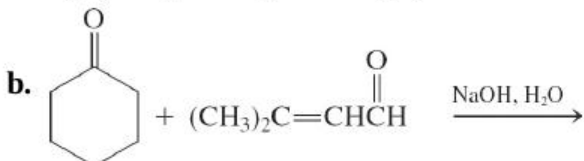
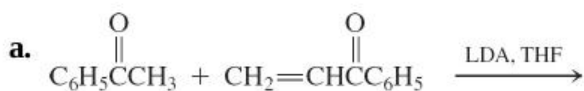


56. In each of the following retrosynthetic disconnections (wavy), show a

reaction or reaction sequence that would make the indicated C—CC—C bonds.



57. Write the products of each of the following room temperature reactions after aqueous work-up.



e. Write the results that you expect from base treatment of the products of reactions (c) and (d) in boiling solvent.